

XUHAN ZHUANG

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EDUCATION

BSc Computer Science

Maastricht University, NL

Sep 2023 - Jun 2026

8.0/10

BSc Economics

Central University of Finance and Economics, CN

Sep 2016 - Jun 2020

72/100

EXPERIENCE

ASML: Software Engineer Intern

Jan 2026 - Present

- Built full-stack platform on **Spring Boot 3**, **React**, and **MySQL** with **RESTful APIs** to replace manual Excel-based workflows, reducing deliverable generation time by **35%**.
- Integrated the platform with existing **Azure** data pipelines across **Databricks**, **Service Bus**, and **Key Vault**; Debugged cross-service data flow issues and shipped fixes with team members.
- Containerized **backend services** with **Docker** and deployed to ASML's internal **Kubernetes** cluster with automated rollouts, service health and performance monitoring.

Stealth Startup: B2B Wholesale, Co-founder

Jan 2022 - Mar 2023

- Co-founded a B2B wholesale venture supplying international e-commerce stores through direct factory sourcing in Fujian, China; owned supplier negotiations, logistics, and end-to-end order fulfillment.

Deloitte: Data Engineer

Oct 2020 - Oct 2021

- Built **ETL pipelines** in **Python**, **Java**, and **Spark** to process and transform large-scale financial datasets into **PostgreSQL** for audit analysis and reporting, reducing manual operation time by 26%.
- Collaborated with IT audit teams to assess database integrity, writing **SQL** queries and developing automated data validation frameworks in **Python** to ensure **100%** accuracy in financial reporting outputs.

PWC: Data Analytics Consulting Intern

Jul 2019 – Oct 2019

- Built **Power BI** dashboards and automated data workflows in **Python** and **SQL** for financial-services consulting engagements, reducing analysis time by **30%** across 5 client projects.

PROJECTS

Folio Multi-Agent Article Generator: Spring Boot 3 + Vue 3 + Spring AI Alibaba ([Link](#) / [GitHub](#))

- Built a full-stack platform to generate articles with LLMs, orchestrating **5 agents** across multi-stage workflows with user intervention, real-time streaming via **SSE** and **concurrent** image generation across 6 providers.
- Persisted to **Cloudflare R2**, **Stripe** VIP payment, deployed via **Docker** and **GitHub Actions**.

Prefix-Aware LLM Cache Router: Go + vLLM + llama.cpp ([GitHub](#))

- Built a **cache-aware router in Go** using per-worker **radix trees** with LRU eviction; achieved **95%** upstream KV-cache hits ($4\times$ A100) and **55%** lower p95 latency vs the best baseline (Apple Silicon).
- Hardened with per-backend **circuit breakers**, **Prometheus** metrics, **SSE** streaming; **97%** test coverage.

Qwen2.5 Text-to-SQL Distillation: LoRA + MLX + 4-bit quantization ([Link](#) / [GitHub](#))

- Distilled GPT-4o-mini's Text-to-SQL into **five Qwen2.5 students** (0.5B to 7B); the 7B reaches **75.0%** on Spider dev vs the teacher's **80.1%**, and the 4-bit 1.5B runs in **847 MB** at **1.16s/query** on a laptop.
- Trained on **Apple Silicon** (mlx-lm) and **4x A100** (transformers + peft) with **LoRA** adapters; filtered teacher samples by **SQL execution match** against gold; deployed as a live **Gradio** demo on **HuggingFace Spaces**.

OPEN SOURCE CONTRIBUTIONS

Merged PRs across **PyTorch**, **React**, **Spring Boot**, **Spring AI**, **Vercel AI SDK**, **langchain4j**, and **JetBrains/Koog** for agentic frameworks and backend infrastructure. ([Link](#))

TECHNICAL SKILLS

Programming: Java, Python, Go, Kotlin, TypeScript/JavaScript, C++, C, Bash, SQL

Backend Development: Spring Boot, REST APIs, Microservices, Distributed Systems, CI/CD, Spark, Kafka, gRPC

Database: MySQL, PostgreSQL, Redis, MongoDB, TiDB

Cloud & DevOps: Docker, Kubernetes, Git, Linux, Azure (AKS, Bicep), Maven/Gradle, AWS, Terraform